Project Synopsis on

LOST AND FOUND MANAGEMENT SYSTEM

Database Management Systems (UCS310)

THAPAR INSTITUTE
OF ENGINEERING & TECHNOLOGY
(Deemed to be University)

Submitted By :
Laksh Khatri (1024030345)
HarshVardhan Singh Naruka (1024030341)
Abhiraj Singh (1024030350)

Submitted to :
Dr. Amrita Dahiya

1. Introduction

In educational institutions, workplaces, and public spaces, the loss of personal belongings is a frequent occurrence. Most organizations still depend on traditional manual systems such as registers or verbal communication to manage lost and found items. These methods are time-consuming, inefficient, and prone to errors.
With the increasing volume of users and items, manual systems fail to provide proper tracking, verification, and accountability. A Database Management System (DBMS) offers a structured and reliable solution by storing data in an organized manner and enabling fast retrieval.
The Lost and Found Management System is designed using Oracle SQL to provide a centralized, secure, and efficient platform for managing lost and found records. The system improves transparency, reduces manual effort, and enhances the chances of successful item recovery.

2. Problem Statement

The existing lost-and-found systems in many institutions are largely manual and unstructured. These systems suffer from several limitations such as poor record maintenance, duplication of data, lack of ownership verification, and delayed processing of claims.
Due to the absence of a centralized database, it becomes difficult to match lost items with found items accurately. Furthermore, there is no systematic mechanism for monitoring claim requests and approvals.
The objective of this project is to overcome these limitations by developing a database-driven solution that provides secure storage, systematic matching, and controlled verification of lost and found items.

3. Objectives of the Project

The main objectives of this project are:

- To design the database using Entity-Relationship (ER) modeling.
- To convert the ER model into relational database tables.
- To apply normalization up to Third Normal Form (3NF).
- To implement database structures using SQL.
- To develop PL/SQL programs for automation and validation.
- To enforce data integrity using primary keys, foreign keys, and constraints.
- To ensure reliability through transaction management.

These objectives aim to enhance data reliability, system efficiency and user satisfaction.

4. Scope of the Project

The scope of this project is limited to the design and implementation of the backend database system.
The system covers the following areas:

- Management of user records.
- Storage of lost and found item details.
- Processing of claim requests.
- Verification by administrators.
- Maintenance of item status.

The system supports two major user roles:

- Users: Report and claim items.
- Administrators: Verify and approve claims.

The project does not include a complete graphical user interface and mainly focuses on database operations and backend functionality.

5. Proposed System Description

The proposed Lost and Found Management System is a centralized database system that stores all relevant information regarding users, items and claims.

- Users register and enter their personal details.
- A user reports an item as lost or found by submitting relevant information.
- The system stores item details in appropriate tables.
- When a potential match is identified, users submit claim requests.
- Administrators verify the authenticity of claims.
- Once approved, the item is returned to its rightful owner.

All operations are managed using SQL and PL/SQL, ensuring accuracy, consistency and automation.

Key Features

- Centralized data storage
- Automated matching of records
- Secure claim verification
- Role-based access control
- Status tracking of items
- Error handling using PL/SQL
- High data consistency

6. Database Design

6.1 Entity-Relationship Diagram

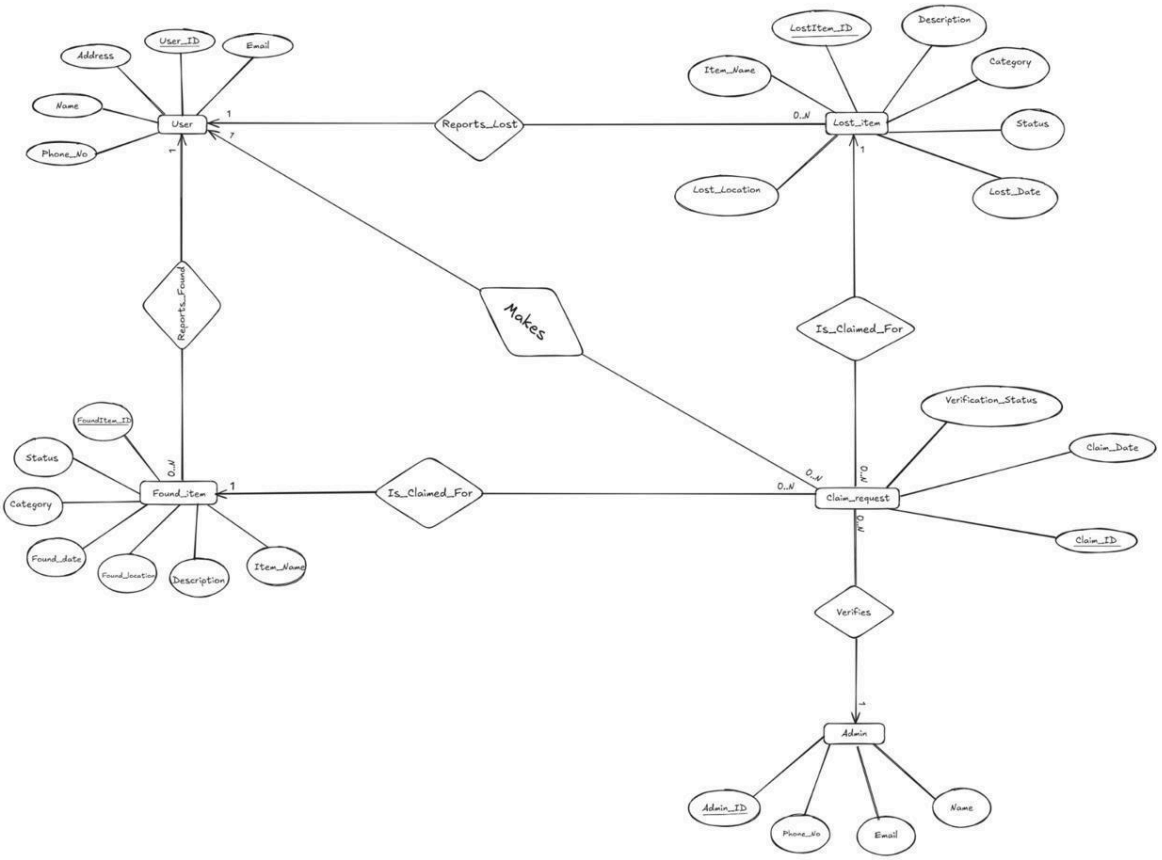
The database design is based on an Entity-Relationship (ER) model. The following entities are identified:

- User
- Lost_Item

- Found_Item
- Claim_Request
- Admin

Each entity contains relevant attributes such as identification numbers, names, contact details, locations, dates, and status.
The relationships among these entities include reporting of items, claim requests, and verification. Proper cardinality and participation constraints are defined to ensure data integrity.
The ER diagram represents one-to-many and many-to-many relationships with suitable constraints.

ENTITY-RELATIONSHIP DIAGRAM



6.2 Relational Schema

The ER diagram is converted into relational tables as follows:

- User(User_ID, Name, Email, Phone_No, Address)
- Lost_Item(LostItem_ID, User_ID, Item_Name, Category, Description, Lost_Location, Lost_Date, Status)
- Found_Item(FoundItem_ID, User_ID, Item_Name, Category, Description, Found_Location, Found_Date, Status)

- Claim_Request(Claim_ID, LostItem_ID, FoundItem_ID, Claimed_By, Claim_Date, Verification_Status)
- Admin(Admin_ID, Name, Email, Phone_No)

Primary keys uniquely identify each record, while foreign keys maintain referential integrity.

6.3 Key relationships

- User — Lost_Item
  One User can report multiple Lost_Items (1:N).
  Each Lost_Item must be reported by exactly one User.
- User — Found_Item
  One User can report multiple Found_Items (1:N).
- User — Claim_Request
  One User can submit multiple Claim_Requests (1:N).
- Admin — Claim_Request
  One Admin can verify multiple Claim_Requests (1:N).
- Lost_Item — Found_Item
  Many Lost_Items may match many Found_Items (M:N).

This relationship is implemented through the Claim_Request table.

6.4 Constraints

Participation Constraints:

- Every Lost_Item must be associated with exactly one User.
- A Claim_Request must be submitted by a registered User.
- Verification of a Claim_Request depends on Admin approval.
- Lost_Items and Found_Items may remain unclaimed (partial participation).

Integrity Constraints:

- Primary Keys uniquely identify each record.
- Foreign Keys maintain referential integrity between related tables.
- Email attributes are defined as UNIQUE.
- Status attributes follow controlled values such as (‘Unclaimed’, ‘Claimed’, ‘Pending’).

7. Normalization

Normalization is applied to minimize redundancy and dependency anomalies.
The normalization process includes:

- First Normal Form (1NF): Removal of repeating groups and multivalued attributes.
- Second Normal Form (2NF): Elimination of partial dependencies.
- Third Normal Form (3NF): Removal of transitive dependencies.

All tables are normalized up to 3NF, ensuring efficient storage and consistent data.

8. Database Implementation

8.1 SQL Implementation

SQL is used to implement database operations, including:

- Data Definition Language (DDL): CREATE, ALTER, DROP
- Data Manipulation Language (DML): INSERT, UPDATE, DELETE
- Data Retrieval using SELECT with:
    - Joins
    - Subqueries
    - Aggregate functions
    - GROUP BY and HAVING clauses
    - Views

These commands enable effective data management and retrieval.

8.2 PL/SQL Components

PL/SQL is used to enhance the functionality of the system through:

- Stored procedures for claim processing
- Functions for validation
- Triggers for automatic status updates
- Cursors for record traversal
- Exception handling for error control

These components improve automation and system reliability.

9. Transaction Management and Concurrency

Transaction control commands such as COMMIT, ROLLBACK, and SAVEPOINT are used to maintain database consistency.
The system follows ACID properties:

- Atomicity
- Consistency
- Isolation
- Durability

Basic concurrency control mechanisms are considered to prevent data conflicts.

10. Tools and Technologies Used

- Database Management System: Oracle Database
- Programming Language: SQL and PL/SQL
- Interface Tool: Oracle SQL Developer
- Operating System: Windows
- Demonstration: Basic frontend

These tools support reliable development and testing.

11. Expected Outcomes

The expected outcomes of the project are:

- Successful development of a normalized database system

- Efficient storage and retrieval of records
- Automated processing of claim requests
- Improved data security and integrity
- Reduction in manual workload
- Faster recovery of lost items
- Increased transparency and accountability